

连续傅里叶变换		连续拉普拉斯变换(单边)		离散 Z 变换(单边)		离散傅里叶变换	
$F(j\omega) = \int_{-\infty}^{\infty} f(t)e^{-j\omega t} dt$ $f(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} F(j\omega)e^{j\omega t} d\omega$		$F(s) = \int_0^{\infty} f(t)e^{-st} dt$ $f(t) = \frac{1}{2\pi j} \int_{\sigma-j\infty}^{\sigma+j\infty} F(s)e^{st} ds$		$F(z) = \sum_{k=0}^{\infty} f(k)z^{-k}$ $f(k) = \frac{1}{2\pi j} \oint_L F(z)z^{k-1} dz, k \geq 0$		$F(e^{j\theta}) = \sum_{k=-\infty}^{\infty} f(k)e^{-j\theta k}$ $f(k) = \frac{1}{2\pi} \int_{-\pi}^{\pi} F(e^{j\theta})e^{j\theta k} d\theta$	
线性	$af_1(t) + bf_2(t) \leftrightarrow aF_1(j\omega) + bF_2(j\omega)$	线性	$af_1(t) + bf_2(t) \leftrightarrow aF_1(s) + bF_2(s)$	线性	$af_1(k) + bf_2(k) \leftrightarrow aF_1(z) + bF_2(z)$	线性	$af_1(k) + bf_2(k) \leftrightarrow aF_1(e^{j\theta}) + bF_2(e^{j\theta})$
时移	$f(t \pm t_0) \leftrightarrow e^{\pm j\omega t_0} F(j\omega)$	时移	$f(t \pm t_0) \leftrightarrow e^{\pm st_0} F(s)$	时移	$f(k \pm m) \leftrightarrow z^{\pm m} F(z)$ (双边)	时移	$f(k \pm m) \leftrightarrow e^{\pm j\theta m} F(e^{j\theta})$
频移	$e^{\pm j\omega_0 t} f(t) \leftrightarrow F(j(\omega \mp \omega_0))$	频移	$e^{\pm st_0} f(t) \leftrightarrow F(s \mp s_0)$	频移	$e^{\pm j\omega_0 k} f(k) \leftrightarrow F(e^{\pm j\theta_0} z)$ (尺度变换)	频移	$e^{\pm j\theta_0 k} f(k) \leftrightarrow F(e^{j(\theta \mp \theta_0)})$
尺度变换	$f(at+b) \leftrightarrow \frac{1}{ a } e^{j\frac{b\omega}{a}} F(j\frac{\omega}{a})$	尺度变换	$f(at+b) \leftrightarrow \frac{1}{ a } e^{\frac{bs}{a}} F(\frac{s}{a})$	尺度变换	$a^k f(k) \leftrightarrow F(\frac{z}{a})$	尺度变换	$f_{(n)}(k) = \begin{cases} f(k/n) \\ 0 \end{cases} \leftrightarrow F(e^{jn\theta})$
反转	$f(-t) \leftrightarrow F(-j\omega)$	反转	$f(-t) \leftrightarrow F(-s)$	反转	$f(-k) \leftrightarrow F(z^{-1})$ (仅限双边)	反转	$f(-k) \leftrightarrow F(e^{-j\theta})$
时域卷积	$f_1(t) * f_2(t) \leftrightarrow F_1(j\omega)F_2(j\omega)$	时域卷积	$f_1(t) * f_2(t) \leftrightarrow F_1(s)F_2(s)$	时域卷积	$f_1(k) * f_2(k) \leftrightarrow F_1(z)F_2(z)$	时域卷积	$f_1(k) * f_2(k) \leftrightarrow F_1(e^{j\theta})F_2(e^{j\theta})$
频域卷积	$f_1(t)f_2(t) \leftrightarrow \frac{1}{2\pi} F_1(j\omega) * F_2(j\omega)$	时域微分	$f'(t) \leftrightarrow sF(s) - f(0_-)$ $f''(t) \leftrightarrow s^2 F(s) - sf'(0_-) - f''(0_-)$	时域差分	$f(k-1) \leftrightarrow z^{-1}F(z) + f(-1)$ $f(k-2) \leftrightarrow z^{-2}F(z) + z^{-1}f(-1) + f(-2)$ $f(k+1) \leftrightarrow zF(z) - zf(0)$ $f(k+2) \leftrightarrow z^2F(z) - z^2f(0) - zf(1)$	频域卷积	$f_1(k)f_2(k) \leftrightarrow \frac{1}{2\pi} \int_{-\pi}^{\pi} F_1(e^{j\psi})F_2(e^{j(\theta-\psi)}) d\psi$
时域微分	$f'(t) \leftrightarrow j\omega F(j\omega)$	频域微分	$f(t)/(-t)^n \leftrightarrow j \frac{d^n F(s)}{ds^n}$	Z域微分	$kf(k) \leftrightarrow -z \frac{dF(z)}{dz}$	时域差分	$f(k) - f(k-1) \leftrightarrow (1 - e^{-j\theta})F(e^{j\theta})$
频域微分	$tf(t) \leftrightarrow (-j)^n f(t) \leftrightarrow j \frac{d^n F(j\omega)}{d\omega^n}$	S域微分	$f(t)/(-t)^n \leftrightarrow -F'(s) \frac{d^n F(s)}{ds^n}$	Z域微分	$kf(k) \leftrightarrow -z \frac{dF(z)}{dz}$	频域微分	$kf(k) \leftrightarrow j \frac{dF(e^{j\theta})}{d\theta}$
时域积分	$\int_{-\infty}^t f(x)dx, f(-\infty)=0 \leftrightarrow \frac{F(j\omega)}{j\omega} + \pi f(0)\delta(\omega)$	时域积分	$\int_{-\infty}^t f(x)dx \leftrightarrow \frac{F(s)}{s} + \frac{f(-1)(0_-)}{s}$	时域积分	$\int_{-\infty}^t f(x)dx \leftrightarrow \frac{F(s)}{s} + \frac{f(-1)(0_-)}{s}$	时域积分	$\int_{-\infty}^t f(k) \leftrightarrow \frac{F(e^{j\theta})}{1 - e^{-j\theta}} + \pi F(e^{j\theta}) \sum_{k=-\infty}^{\infty} \delta(\theta - 2\pi k)$
频域积分	$\pi f(0) + \frac{f(t)}{(-jt)} \leftrightarrow \int_{-\infty}^{\infty} F(j\tau) d\tau, F(-\infty)=0$	S域积分	$\frac{f(t)}{t} \leftrightarrow \int_s^{\infty} F(\eta) d\eta$	Z域积分	$\frac{f(k)}{k+m} \leftrightarrow z^m \int_z^{\infty} \frac{F(\eta)}{\eta^{m+1}} d\eta$	频域积分	$f(0) = \lim_{z \rightarrow \infty} F(z), f(1) = \lim_{z \rightarrow \infty} [zF(z) - zf(0)]$
对称	$F(j\omega) \leftrightarrow 2\pi f(-\omega)$	初值	$f(0_+) = \lim_{s \rightarrow \infty} sF(s), F(s) \text{ 为真分式}$	初值	$f(M) = \lim_{z \rightarrow \infty} z^M F(z)$ (右边信号), $f(M+1) = \lim_{z \rightarrow \infty} [z^{M+1} F(z) - zf(M)]$	帕斯瓦尔	$\sum_{k=-\infty}^{\infty} f(k) ^2 = \frac{1}{2\pi} \int_{-\pi}^{\pi} F(e^{j\theta}) ^2 d\theta$
帕斯瓦尔	$E = \int_{-\infty}^{\infty} f(t) ^2 dt = \frac{1}{2\pi} \int_{-\infty}^{\infty} F(j\omega) ^2 d\omega$	终值	$f(\infty) = \lim_{s \rightarrow 0} sF(s), s=0 \text{ 在收敛域内}$	终值	$f(\infty) = \lim_{z \rightarrow 1} (z-1)F(z)$ (右边信号)	帕斯瓦尔	



夸克扫描王

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常用连续傅里叶变换、拉普拉斯变换、Z变换对一览表

连续傅里叶变换对		拉普拉斯变换对 (单边)		Z变换对 (单边)			
$F(j\omega) = \int_{-\infty}^{\infty} f(t)e^{-j\omega t} dt$		$F(s) = \int_0^{\infty} f(t)e^{-st} dt$		$F(z) = \sum_{k=0}^{\infty} f(k)z^{-k}$			
函数 $f(t)$	傅里叶变换 $F(j\omega)$	函数 $f(t)$	拉普拉斯变换 $F(s)$	函数 $f(k), k \geq 0$	象函数	函数 $f(k), k \geq 0$	象函数
$\delta(t)/1$	$\sqrt{2\pi}\delta(\omega)$	$\delta(t)$	1	$\delta(k)$	1	$\delta(k-m), m \geq 0$	z^{-m}
$\delta'(t)/\delta^{(n)}(t)$	$j\omega/(j\omega)^n$	$\delta'(t)$	s	1	$\frac{z}{z-1}$	$\varepsilon(k-m), m \geq 0$	$\frac{z^{-m}}{z-1}$
$\varepsilon(t)$	$\frac{1}{j\omega} + \pi\delta(\omega)$	$\varepsilon(t)$	$\frac{1}{s}$	$\varepsilon(k)$	$\frac{z}{z-1}$	$k^2\varepsilon(k)$	$\frac{z^2+z}{(z-1)^3}$
$t\varepsilon(t)$	$j\pi\delta'(\omega) - \frac{1}{\omega^2}$	$t\varepsilon(t)/t^n\varepsilon(t)$	$\frac{1}{s} / \frac{n!}{s^{n+1}}$	$k\varepsilon(k)$	$\frac{z}{(z-1)^2}$	$(k+1)a^k\varepsilon(k)$	$\frac{z^2}{(z-a)^2}$
$e^{-\alpha t}\varepsilon(t)/te^{-\alpha t}\varepsilon(t), \alpha > 0$	$\frac{1}{\alpha + j\omega} / \frac{1}{(\alpha + j\omega)^2}$	$e^{-\alpha t}\varepsilon(t)/te^{-\alpha t}\varepsilon(t)$	$\frac{1}{s+\alpha} / \frac{1}{(s+\alpha)^2}$	$a^k\varepsilon(k)$	$\frac{z}{z-a}$	$ka^{k-1}\varepsilon(k)$	$\frac{z}{(z-a)^2}$
$\cos(\omega_0 t)$ $\sin(\omega_0 t)$	$\pi[\delta(\omega+\omega_0) + \delta(\omega-\omega_0)]$ $j\pi[\delta(\omega+\omega_0) - \delta(\omega-\omega_0)]$	$\cos(\beta t)\varepsilon(t)$	$\frac{s}{s^2 + \beta^2}$	$e^{a^k}\varepsilon(k)$	$\frac{z}{z-e^a}$	$ka^k\varepsilon(k)$	$\frac{az}{(z-a)^2}$
$\frac{1}{t}$	$-j\pi \operatorname{sgn}(\omega)$	$\sin(\beta t)\varepsilon(t)$	$\frac{\beta}{s^2 + \beta^2}$	$e^{j\beta k}\varepsilon(k)$	$\frac{z}{z-e^{j\beta}}$	$k^2a^k\varepsilon(k)$	$\frac{az^2+a^2z}{(z-a)^3}$
$ t $	$-\frac{2}{\omega^2}$	$\cosh(\beta t)\varepsilon(t)$	$\frac{s}{s^2 - \beta^2}$	$\frac{a^k - (-a)^k}{2a}\varepsilon(k)$	$\frac{z}{z^2 - a^2}$	$\frac{a^k + (-a)^k}{2a}\varepsilon(k)$	$\frac{z^2}{z^2 - a^2}$
$e^{\pm j\omega_0 t}$	$2\pi\delta(\omega \mp \omega_0)$	$\sinh(\beta t)\varepsilon(t)$	$\frac{\beta}{s^2 - \beta^2}$	$\frac{k(k-1)}{2}\varepsilon(k)$	$\frac{z}{(z-1)^3}$	$\frac{(k+1)k}{2}\varepsilon(k)$	$\frac{z^2}{(z-1)^3}$
$e^{-\alpha t}\cos(\beta t)\varepsilon(t)$	$\frac{j\omega + \alpha}{(j\omega + \alpha)^2 + \beta^2}$	$e^{-\alpha t}\cos(\beta t)\varepsilon(t)$	$\frac{s + \alpha}{(s + \alpha)^2 + \beta^2}$	$\frac{a^k - b^k}{a-b}\varepsilon(k)$	$\frac{z}{(z-a)(z-b)}$	$\frac{a^{k+1} - b^{k+1}}{a-b}\varepsilon(k)$	$\frac{z^2}{(z-a)(z-b)}$
$e^{-\alpha t}\sin(\beta t)\varepsilon(t)$	$\frac{\beta}{(j\omega + \alpha)^2 + \beta^2}$	$e^{-\alpha t}\sin(\beta t)\varepsilon(t)$	$\frac{\beta}{(s + \alpha)^2 + \beta^2}$	$\cos(\beta k)\varepsilon(k)$	$\frac{z(z - \cos \beta)}{z^2 - 2z \cos \beta + 1}$	$\sin(\beta k)\varepsilon(k)$	$\frac{z \sin \beta}{z^2 - 2z \cos \beta + 1}$



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双边拉普拉斯变换与双边 Z 变换对一览表

双边拉普拉斯变换对		双边 Z 变换对	
$F(s) = \int_{-\infty}^{\infty} f(t)e^{-st} dt$		$F(z) = \sum_{k=-\infty}^{\infty} f(k)z^{-k}$	
函数	象函数 $F(s)$ 和收敛域	函数	象函数 $F(z)$ 和收敛域
$\delta(t)$	1, 整个 S 平面	$\delta(k)$	1, 整个 Z 平面
$\delta^{(n)}(t)$	s^n , 有限 S 平面	$\Delta^n \delta(k)$	$\frac{z^n}{(z-1)^n}, z > 0$
$\varepsilon(t)$	$\frac{1}{s}, \text{Re}\{s\} > 0$	$\varepsilon(k)$	$\frac{z}{z-1}, z > 1$
$t\varepsilon(t)$	$\frac{1}{s^2}, \text{Re}\{s\} > 0$	$(k+1)\varepsilon(k)$	$\frac{z^2}{(z-1)^2}, z > 1$
$\frac{t^{n-1}}{(n-1)!}\varepsilon(t)$	$\frac{1}{s^n}, \text{Re}\{s\} > 0$	$\frac{(k+n-1)!}{k!(n-1)!}\varepsilon(k)$	$\frac{z^n}{(z-1)^n}, z > 1$
$-\varepsilon(-t)$	$\frac{1}{s}, \text{Re}\{s\} < 0$	$-\varepsilon(-k-1)$	$\frac{z}{z-1}, z < 1$
$-t\varepsilon(-t)$	$\frac{1}{s^2}, \text{Re}\{s\} < 0$	$-(k+1)\varepsilon(-k-1)$	$\frac{z^2}{(z-1)^2}, z < 1$
$-\frac{t^{n-1}}{(n-1)!}\varepsilon(-t)$	$\frac{1}{s^n}, \text{Re}\{s\} < 0$	$-\frac{(k+n-1)!}{k!(n-1)!}\varepsilon(-k-1)$	$\frac{z^n}{(z-1)^n}, z < 1$
$e^{-at}\varepsilon(t)$	$\frac{1}{s+a}, \text{Re}\{s\} > \text{Re}\{-a\}$	$a^k \varepsilon(k)$	$\frac{z}{z-a}, z > a $
$te^{-at}\varepsilon(t)$	$\frac{1}{(s+a)^2}, \text{Re}\{s\} > \text{Re}\{-a\}$	$(n+1)a^n \varepsilon(k)$	$\frac{z^2}{(z-a)^2}, z > a $
$\frac{t^{n-1}}{(n-1)!}e^{-at}\varepsilon(t)$	$\frac{1}{(s+a)^n}, \text{Re}\{s\} > \text{Re}\{-a\}$	$\frac{(k+n-1)!}{k!(n-1)!}a^n \varepsilon(k)$	$\frac{z^n}{(z-a)^n}, z > a $
$-e^{-at}\varepsilon(-t)$	$\frac{1}{s+a}, \text{Re}\{s\} < \text{Re}\{-a\}$	$-a^k \varepsilon(-k-1)$	$\frac{z}{z-a}, z < a $
$-\frac{t^{n-1}}{(n-1)!}e^{-at}\varepsilon(-t)$	$\frac{1}{(s+a)^n}, \text{Re}\{s\} < \text{Re}\{-a\}$	$-\frac{(k+n-1)!}{k!(n-1)!}a^n \varepsilon(-k-1)$	$\frac{z^n}{(z-a)^n}, z < a $
$\cos(\beta t)\varepsilon(t)$	$\frac{s}{s^2 + \beta^2}, \text{Re}\{s\} > 0$	$\cos(\beta k)\varepsilon(k)$	$\frac{z^2 - z \cos \beta}{z^2 - 2z \cos \beta + 1}$
$\sin(\beta t)\varepsilon(t)$	$\frac{\beta}{s^2 + \beta^2}, \text{Re}\{s\} > 0$	$\sin(\beta k)\varepsilon(k)$	$\frac{z \sin \beta}{z^2 - 2z \cos \beta + 1}$
$e^{-\alpha t} \cos(\beta t)\varepsilon(t)$	$\frac{s + \alpha}{(s + \alpha)^2 + \beta^2}, \text{Re}\{s\} > \text{Re}\{-\alpha\}$	$a^k \cos(\beta k)\varepsilon(k)$	$\frac{z^2 - za \cos \beta}{z^2 - 2za \cos \beta + 1}$
$e^{-\alpha t} \sin(\beta t)\varepsilon(t)$	$\frac{\beta}{(s + \alpha)^2 + \beta^2}, \text{Re}\{s\} > \text{Re}\{-\alpha\}$	$a^k \sin(\beta k)\varepsilon(k)$	$\frac{za \sin \beta}{z^2 - 2za \cos \beta + 1}$
$e^{-a t }, \text{Re}\{a\} > 0$	$\frac{-2a}{s^2 - a^2}, \text{Re}\{a\} > \text{Re}\{s\} > \text{Re}\{-a\}$	$a^{ k }, a < 1$	$\frac{(a^2 - 1)z}{(z-a)(az-1)}, a < z < \frac{1}{ a }$
$e^{-a t } \text{sgn}(t), \text{Re}\{a\} > 0$	$\frac{2s}{s^2 - a^2}, \text{Re}\{a\} > \text{Re}\{s\} > \text{Re}\{-a\}$	$a^{ k } \text{sgn}, a < 1$	$\frac{a(z^2 - z)}{(z-a)(az-1)}, a < z < \frac{1}{ a }$



卷积积分一览表

$f_1(t) * f_2(t) = \int_{-\infty}^{\infty} f_1(\tau) f(t-\tau) d\tau$					
$f_1(t)$	$f_2(t)$	$f_1(t) * f_2(t)$	$f_1(t)$	$f_2(t)$	$f_1(t) * f_2(t)$
$f(t)$	$\delta'(t)$	$f'(t)$	$f(t)$	$\delta(t)$	$f(t)$
$f(t)$	$\varepsilon(t)$	$\int_{-\infty}^t f(\lambda) d\lambda$	$\varepsilon(t)$	$\varepsilon(t)$	$t\varepsilon(t)$
$e^{-\alpha t} \varepsilon(t)$	$\varepsilon(t)$	$\frac{1}{\alpha} (1 - e^{-\alpha t}) \varepsilon(t)$	$\varepsilon(t)$	$t\varepsilon(t)$	$\frac{1}{2} t^2 \varepsilon(t)$
$e^{-\alpha_1 t} \varepsilon(t)$	$e^{-\alpha_2 t} \varepsilon(t)$	$\frac{1}{\alpha_2 - \alpha_1} (e^{-\alpha_1 t} - e^{-\alpha_2 t}) \varepsilon(t), \alpha_1 \neq \alpha_2$	$e^{-\alpha t} \varepsilon(t)$	$e^{-\alpha t} \varepsilon(t)$	$t e^{-\alpha t} \varepsilon(t)$
$t e^{-\alpha_1 t} \varepsilon(t)$	$e^{-\alpha_2 t} \varepsilon(t)$	$\left[\frac{(\alpha_2 - \alpha_1)t - 1}{(\alpha_2 - \alpha_1)^2} e^{-\alpha_1 t} + \frac{1}{(\alpha_2 - \alpha_1)^2} e^{-\alpha_2 t} \right] \varepsilon(t)$ $\alpha_2 \neq \alpha_1$	$t\varepsilon(t)$	$e^{-\alpha t} \varepsilon(t)$	$\left(\frac{\alpha t - 1}{\alpha^2} + \frac{1}{\alpha^2} e^{-\alpha t} \right) \varepsilon(t)$
$e^{-\alpha_1 t} \cos(\beta t + \theta) \varepsilon(t)$	$e^{-\alpha_2 t} \varepsilon(t)$	$\left[\frac{\cos(\beta t + \theta - \varphi)}{\sqrt{(\alpha_2 - \alpha_1)^2 + \beta^2}} e^{-\alpha_1 t} - \frac{\cos(\theta - \varphi)}{\sqrt{(\alpha_2 - \alpha_1)^2 + \beta^2}} e^{-\alpha_2 t} \right] \varepsilon(t)$ $\varphi = \arctan\left(\frac{\beta}{\alpha_2 - \alpha_1}\right)$	$t e^{-\alpha t} \varepsilon(t)$	$e^{-\alpha t} \varepsilon(t)$	$\frac{1}{2} t^2 e^{-\alpha t} \varepsilon(t)$

卷积和一览表

$f_1(i) * f_2(i) = \sum_{i=-\infty}^{\infty} f_1(i) f(k-i)$					
$f_1(t)$	$f_2(t)$	$f_1(t) * f_2(t)$	$f_1(t)$	$f_2(t)$	$f_1(t) * f_2(t)$
$f(k)$	$\delta(k)$	$f(k)$	$f(k)$	$\varepsilon(k)$	$\sum_{i=-\infty}^k f(i)$
$\varepsilon(k)$	$\varepsilon(k)$	$(k+1)\varepsilon(k)$	$k\varepsilon(k)$	$\varepsilon(k)$	$\frac{1}{2}(k+1)k\varepsilon(k)$
$a^k \varepsilon(k)$	$\varepsilon(k)$	$\frac{1-a^{k+1}}{1-a} \varepsilon(k), a \neq 0$	$a_1^k \varepsilon(k)$	$a_2^k \varepsilon(k)$	$\frac{a_1^{k+1} - a_2^{k+1}}{a_1 - a_2} \varepsilon(k), a_1 \neq a_2$
$a^k \varepsilon(k)$	$a^k \varepsilon(k)$	$(k+1)a^k \varepsilon(k)$	$k\varepsilon(k)$	$a^k \varepsilon(k)$	$\frac{k}{1-a} \varepsilon(k) + \frac{a(a^k - 1)}{(1-a)^2} \varepsilon(k)$
$k\varepsilon(k)$	$k\varepsilon(k)$	$\frac{1}{6}(k+1)k(k-1)\varepsilon(k)$	$a_1^k \cos(\beta k + \theta) \varepsilon(k)$	$a^k \varepsilon(k)$	$\frac{a_1^{k+1} \cos[\beta(k+1) + \theta - \varphi] - a_2^{k+1} \cos(\theta - \varphi)}{\sqrt{a_1^2 + a_2^2 - a_1 a_2 \cos \beta}} \varepsilon(k)$ $\varphi = \arctan\left[\frac{a_1 \sin \beta}{a_1 \cos \beta - a_2}\right]$

关于 $\delta(t)$ 、 $\delta(k)$ 函数公式一览表

$f(t)\delta(t) = f(0)\delta(t)$	$f(t)\delta(t-t_0) = f(t_0)\delta(t-t_0)$	$\delta(-t) = \delta(t)/\delta'(-t) = -\delta'(t)$	$f(t)\delta'(t) = f(0)\delta'(t) - f'(0)\delta(t)$
$\int_{-\infty}^{\infty} f(t)\delta(t)dt = f(0)$	$\int_{-\infty}^{\infty} f(t)\delta(t-t_0)dt = f(t_0)$	$\delta[f(t)] = \sum_{i=1}^n \frac{1}{ f'(t_i) } \delta(t-t_i)$	$\int_{-\infty}^{\infty} f(t)\delta^{(n)}(t)dt = (-1)^n f^{(n)}(0)$
$\delta(at) = \frac{1}{ a } \delta(t)$	$\int_{-\infty}^{\infty} \delta(t)dt = 1 / \int_{-\infty}^t \delta(\tau)d\tau = \varepsilon(t)$	$\int_{-\infty}^{\infty} \delta'(t)dt = 0 / \int_{-\infty}^t \delta'(\tau)d\tau = \delta(t)$	$f(t)\delta'(t-t_0) = f(t_0)\delta'(t-t_0) - f'(t_0)\delta(t-t_0)$



常用的连续傅里叶变换对及其对偶关系					
连续傅里叶变换对			相对偶的连续傅里叶变换对		
重 要	连续时间函数 $f(t)$	傅里叶变换 $F(\omega)$	连续时间函数 $f(t)$	傅里叶变换 $F(\omega)$	重 要
✓	$\delta(t)$	1	1	$2\pi\delta(\omega)$	✓
✓	$\frac{d}{dt}\delta(t)$	$j\omega$	t	$j2\pi\frac{d}{d\omega}\delta(\omega)$	
	$\frac{d^k}{dt^k}\delta(t)$	$(j\omega)^k$	t^k	$2\pi j^k\frac{d^k}{d\omega^k}\delta(\omega)$	
✓	$u(t)$	$\frac{1}{j\omega} + \pi\delta(\omega)$	$\frac{1}{2}\delta(t) - \frac{1}{j2\pi}$	$u(\omega)$	
	$tu(t)$	$j\pi\frac{d}{d\omega}\delta(\omega) - \frac{1}{\omega^2}$			
	$\text{sgn}(t) = \begin{cases} 1, t > 0 \\ -1, t < 0 \end{cases}$	$\frac{2}{j\omega}$	$\frac{1}{\pi}, t \neq 0$	$F(\omega) = \begin{cases} -j, \omega > 0 \\ j, \omega < 0 \end{cases}$	
✓	$\delta(t-t_0)$	$e^{-j\omega t_0}$	$e^{j\omega_0 t}$	$2\pi\delta(\omega-\omega_0)$	✓
	$\cos\omega_0 t$	$\pi[\delta(\omega+\omega_0) + \delta(\omega-\omega_0)]$	$\delta(t+t_0) + \delta(t-t_0)$	$2\cos\omega t_0$	
	$\sin\omega_0 t$	$j\pi[\delta(\omega+\omega_0) - \delta(\omega-\omega_0)]$	$\delta(t+t_0) - \delta(t-t_0)$	$j2\sin\omega t_0$	
✓	$f(t) = \begin{cases} 1, t < \tau \\ 0, t > \tau \end{cases}$	$\tau \text{Sa}(\frac{\omega\tau}{2})$	$\frac{W}{\pi} \text{Sa}(Wt)$	$F(\omega) = \begin{cases} 1, \omega < W \\ 0, \omega > W \end{cases}$	✓
✓	$f(t) = \begin{cases} 1- t /\tau, t < \tau \\ 0, t > \tau \end{cases}$	$\tau \text{Sa}^2(\frac{\omega\tau}{2})$	$\frac{W}{2\pi} \text{Sa}^2(\frac{Wt}{2})$	$F(\omega) = \begin{cases} 1- \omega /W, \omega < W \\ 0, \omega > W \end{cases}$	
✓	$e^{-at}u(t), \text{Re}\{a\} > 0$	$\frac{1}{a+j\omega}$	$\frac{1}{\tau-jt}$	$2\pi e^{-\tau\omega}u(\omega), \tau > 0$	
	$e^{-a t }, \text{Re}\{a\} > 0$	$\frac{2a}{\omega^2 + a^2}$	$\frac{\tau}{t^2 + \tau^2}$	$\pi e^{-\tau \omega }, \tau > 0$	
✓	$e^{-at}\cos\omega_0 t, \text{Re}\{a\} > 0$	$\frac{a+j\omega}{(a+j\omega)^2 + \omega_0^2}$			
✓	$e^{-at}\sin\omega_0 t, \text{Re}\{a\} > 0$	$\frac{\omega_0}{(a+j\omega)^2 + \omega_0^2}$			
	$te^{-at}u(t), \text{Re}\{a\} > 0$	$\frac{1}{(a+j\omega)^2}$	$\frac{1}{(\tau-jt)^2}, \tau > 0$	$2\pi\omega e^{-\tau\omega}u(\omega)$	
	$\frac{t^{k-1}}{(k-1)!}e^{-at}u(t), \text{Re}\{a\} > 0$	$\frac{1}{(a+j\omega)^k}$			
✓	$\delta_T(t) = \sum_{k=-\infty}^{+\infty}\delta(t-kT)$	$\frac{2\pi}{T}\sum_{k=-\infty}^{+\infty}\delta(\omega-k\frac{2\pi}{T})$			
✓	$e^{-\frac{t^2}{\tau^2}}$	$\sqrt{\pi}\tau e^{-\frac{\omega^2\tau^2}{4}}$			
✓	$[\mu(t+\frac{\tau}{2}) - \mu(t-\frac{\tau}{2})]\cos\omega_0 t$	$\frac{\tau}{2}[Sa(\frac{(\omega+\omega_0)\tau}{2}) + Sa(\frac{(\omega-\omega_0)\tau}{2})]$			
	$\sum_{k=-\infty}^{+\infty}F_k e^{jk\omega_0 t}$	$2\pi\sum_{k=-\infty}^{+\infty}F_k\delta(\omega-k\omega_0)$			



夸克扫描王

极速扫描，就是高效



连续傅里叶变换性质及其对偶关系							
$f(t) = \frac{1}{2\pi} \int_{-\infty}^{+\infty} F(\omega) e^{j\omega t} d\omega$ $f(0) = \frac{1}{2\pi} \int_{-\infty}^{+\infty} F(\omega) d\omega$				$F(\omega) = \int_{-\infty}^{+\infty} f(t) e^{-j\omega t} dt$ $F(0) = \int_{-\infty}^{+\infty} f(t) dt$			
连续傅里叶变换对				相对偶的连续傅里叶变换对			
重 要	名称	连续时间函数 $f(t)$	傅里叶变换 $F(\omega)$	名称	连续时间函数 $f(t)$	傅里叶变换 $F(\omega)$	重 要
✓	线性	$\alpha f_1(t) + \beta f_2(t)$	$\alpha F_1(\omega) + \beta F_2(\omega)$				
✓	尺度比例变换	$f(at), a \neq 0$	$\frac{1}{ a } F\left(\frac{\omega}{a}\right)$				
	对偶性	$f(t)$	$g(\omega)$		$g(t)$	$2\pi f(-\omega)$	✓
✓	时移	$f(t-t_0)$	$F(\omega)e^{-j\omega t_0}$	频移	$f(t)e^{j\omega_0 t}$	$F(\omega-\omega_0)$	✓
	时域微分性质	$\frac{d}{dt} f(t)$	$j\omega F(\omega)$	频域微分性质	$-jtf'(t)$	$\frac{d}{d\omega} F(\omega)$	✓
	时域积分性质	$\int_{-\infty}^t f(\tau) d\tau$	$\frac{F(\omega)}{j\omega} + \pi F(0)\delta(\omega)$	频域积分性质	$\frac{f(t)}{-jt} + \pi f(0)\delta(t)$	$\int_{-\infty}^{\omega} F(\sigma) d\sigma$	
✓	时域卷积性质	$f(t)*h(t)$	$F(\omega)H(\omega)$	频域卷积性质	$f(t)p(t)$	$\frac{1}{2\pi} F(\omega)*P(\omega)$	✓
✓	对称性	$f(-t)$ $f^*(t)$ $f^*(-t)$	$F(-\omega)$ $F^*(-\omega)$ $F^*(\omega)$	奇偶虚实性质	$f(t)$ 是实函数 $f(t) = \text{Od}\{f(t)\}$ $f_e(t) = \text{Ev}\{f(t)\}$	$j\text{Im}\{F(\omega)\}$ $\text{Re}\{F(\omega)\}$	
	希尔伯特变换	$f(t) = f(t)u(t)$	$F(\omega) = R(\omega) + jI(\omega)$ $R(\omega) = I(\omega)*\frac{1}{\pi\omega}$				
✓	时域抽样	$f(t) \sum_{n=-\infty}^{+\infty} \delta(t-nT)$	$\frac{1}{T} \sum_{k=-\infty}^{+\infty} F(\omega - k\frac{2\pi}{T})$	频域抽样	$\frac{1}{\omega_0} \sum_{n=-\infty}^{+\infty} f(t-n\frac{2\pi}{\omega_0})$	$F(\omega) \sum_{k=-\infty}^{+\infty} \delta(\omega - k\omega_0)$	
✓	帕什瓦尔公式	$\int_{-\infty}^{+\infty} f(t) ^2 dt = \frac{1}{2\pi} \int_{-\infty}^{+\infty} F(\omega) ^2 d\omega$					

取反-----取反
共轭----共轭取反
共轭取反----共轭

